# JDBC Quick Reference Guide

Complete Java Database Connectivity (JDBC) CRUD Operations Reference

## CREATE Create Table Operation

```java
import java.sql.*;

class CreateTable {
    public static void main(String[] args) throws Exception {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/college",
            "root", "password");

        Statement st = con.createStatement();

        st.executeUpdate(
            "CREATE TABLE student (id INT, name VARCHAR(30))");

        System.out.println("Table created");

        con.close();
    }
}
```

## INSERT Insert Record Operation

```java
import java.sql.*;

class InsertData {
    public static void main(String[] args) throws Exception {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/college",
            "root", "password");

        Statement st = con.createStatement();

        st.executeUpdate(
            "INSERT INTO student VALUES (1, 'Ravi')");

        System.out.println("Record inserted");

        con.close();
    }
}
```

## SELECT Read / Query Data Operation

```java
import java.sql.*;

class SelectData {
    public static void main(String[] args) throws Exception {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/college",
            "root", "password");

        Statement st = con.createStatement();

        ResultSet rs = st.executeQuery(
            "SELECT * FROM student");

        while (rs.next()) {
            System.out.println(
                rs.getInt("id") + " " +
                rs.getString("name"));
        }

        con.close();
    }
}
```

## UPDATE Update Record Operation

```java
import java.sql.*;

class UpdateData {
    public static void main(String[] args) throws Exception {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/college",
            "root", "password");

        Statement st = con.createStatement();

        int rows = st.executeUpdate(
            "UPDATE student SET name = 'Surya' WHERE id = 1");

        if (rows > 0) {
            System.out.println("Record updated successfully!");
        } else {
            System.out.println("No record found.");
        }

        st.close();
        con.close();
    }
}
```

## DELETE Delete Record Operation (PreparedStatement)

```
import java.sql.*;
import java.util.Scanner;

class DeleteData {
    public static void main(String[] args) throws Exception {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/college",
            "root", "password");

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter student ID to delete: ");
        int id = sc.nextInt();

        PreparedStatement ps = con.prepareStatement(
            "DELETE FROM student WHERE id = ?");

        ps.setInt(1, id);

        int rows = ps.executeUpdate();

        if (rows > 0) {
            System.out.println("Student deleted successfully.");
        } else {
            System.out.println("Student ID not found.");
        }

        ps.close();
        con.close();
        sc.close();
    }
}
```